Pre-analysis Plan (RT

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

- Variables:

bhp_504_q56	position of women: working as women
bhp_505_q56	position of women: working as married women
bhp_506_q56	position of women: when woman earns more than partner, problems
bhp_507_q56	position of women: education of son more important than of daughter
bhp_508_q56	position of women: husband should have the final say

EQUALITY_INDEX=\sum [(bhp_i-av(bhp_i))/sd(bhp_i)]/N

Bhp_506_q56, bhp_507_q56, and hp_508_q56 will be recoded so agreeing means being in favor of greater gender equality.

- Inclusion/exclusion criteria:
  Include individuals >18 years with one child only
  Exclude homosexuals (who could potentially choose their child's gender)
  We will append observations from years 2016 and 2017 and keep only the first time an individual is interviewed (in case a given individual appears more than once)

- Exact analysis:
  EQUALITY_INDEX=alpha+beta*Girl+CFE+X+epsilon
  Girl=1 if gender of the child is female, 0 otherwise.
  CFE=cohort fixed effects
  X=controls (gender of respondent, age of respondent, urban/rural, child's age, married, education, log(income), religious_freq, nationality)
  Education=bhp_15_01-bhp_22_08, to be grouped
  Urban/rural=DB100

Income=log(bhp_38_01)
Religious_freq=bhp_10_09 included as one dummy for each category of frequency
Nationality=pnat_v2 included as one dummy for each nationality
Standard errors: given the lack of availability a Sampling/Geographical Unit identifier, we opt to use robust standard errors

- Interpretation of results:
  Beta will measure the difference in attitudes towards gender equality between single-child parents of daughters compared to single-child parents of sons, accounting for relevant covariates.

H2. Having daughters leads to more left-leaning political preferences.

- Variables:

Plh0004 variable is available for years 2005, 2009, 2014 and 2019:
https://paneldata.org/soep-core/datasets/pl/plh0004

"In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?" Answers range between 0 (extreme left) and 10 (extreme right).

We will construct the following variable:
Left=(-1)plh0004+10
Then we will standardise this variable to have mean 0 and standard deviation of 1.

- Inclusion/exclusion criteria:
  Include individuals >18 years with one child only
  Exclude homosexuals (who could potentially choose their child's gender)
  We will append observations from years 2005, 2009, 2014 and 2019 and keep only the first time an individual is interviewed (in case a given individual appears more than once).

- Exact analysis:
  Left=alpha+beta*Girl+CFE+X+epsilon
  Girl=1 if gender of the child is female, 0 otherwise.
  CFE=cohort fixed effects
  X=controls (gender of respondent, age of respondent, urban/rural, child's age, married, education, log(income), religious_freq, nationality)

Education=bhp_15_01-bhp_22_08, to be grouped
Urban/rural=DB100
Income=log(bhp_38_01)
Religious_freq=bhp_10_09 included as one dummy for each category of frequency
Nationality=pnat_v2 included as one dummy for each nationality
Standard errors: given the lack of availability a Sampling/Geographical Unit identifier, we opt to use robust standard errors

- Interpretation of results:
  Beta will measure the difference (in terms of standard deviations) in leaning towards the political left between single-child parents of daughters and single-child parents of sons, accounting for relevant covariates.

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

- Variables:

We will construct a gender inequality index (GII) in household chores:

GII=abs((H-W)/(H+W))

Using the following variable:
https://paneldata.org/soep-core/datasets/pl/pli0043_h

- Inclusion/exclusion criteria:
  Include individuals >18 years with one child only
  Exclude homosexuals (who could potentially choose their child's gender)
  We will append observations from all years in which the question is available along with controls, and keep only the first time an individual is interviewed (in case a given individual appears more than once).
- Exact analysis:
  Left=alpha+beta*Girl+CFE+X+epsilon
  Girl=1 if gender of the child is female, 0 otherwise.
  CFE=cohort fixed effects
  X=controls (gender of respondent, age of respondent, urban/rural, child's age, married, education, log(income), religious_freq, nationality)
  Education=bhp_15_01-bhp_22_08, to be grouped
  Urban/rural=DB100
  Income=log(bhp_38_01)

Religious_freq=bhp_10_09 included as one dummy for each category of frequency
Nationality=pnat_v2 included as one dummy for each nationality
Standard errors: given the lack of availability a Sampling/Geographical Unit identifier, we opt to use robust standard errors

- Interpretation of results:
  Beta will measure the difference in GII between single-child parents of daughters and single-child parents of sons, accounting for relevant covariates.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

- Variables:

bhp_504_q56	position of women: working as women
bhp_505_q56	position of women: working as married women
bhp_506_q56	position of women: when woman earns more than partner, problems
bhp_507_q56	position of women: education of son more important than of daughter
bhp_508_q56	position of women: husband should have the final say

EQUALITY_INDEX=\sum [(bhp_i-av(bhp_i))/sd(bhp_i)]/N
Bhp_506_q56 bhp_507_q56 hp_508_q56 will be recoded so agreeing means being in favor of greater gender equality.

- Inclusion/exclusion criteria:
  Include individuals >18 years with one child only
  Exclude homosexuals (who could potentially choose their child's gender)
  We will append observations from years 2016 and 2017 and keep only the first time an individual is interviewed (in case a given individual appears more than once)

- Exact analysis:
  EQUALITY_INDEX=alpha+beta_0*Girl+beta_1*Girl*Male_Parent+CFE+X+epsilon

Girl=1 if gender of the child is female, 0 otherwise.
Male_Parent=1 for male parent, and 0 for female parent.
CFE=cohort fixed effects
X=controls (age of respondent, urban/rural, child's age, married, education, log(income), religious_freq, nationality)
Education=bhp_15_01-bhp_22_08, to be grouped
Urban/rural=DB100
Income=log(bhp_38_01)
Religious_freq=bhp_10_09 included as one dummy for each category of frequency
Nationality=pnat_v2 included as one dummy for each nationality
Standard errors: given the lack of availability a Sampling/Geographical Unit identifier, we opt to use robust standard errors

- Interpretation of results:
  Beta_1 will measure the differential effect of having daughters versus sons by male and female single-child parents, accounting for relevant covariates.